FORENSIC ANALYSIS REPORT

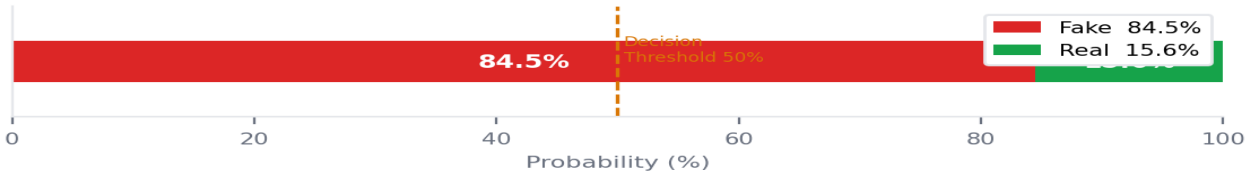
Acoustic Deepfake Detection · Powered by WavLM + Integrated Gradients + SHAP

VERDICT: DEEPFAKE / SYNTHETIC CONTENT DETECTED

WavLM neural network detected AI-generated content fingerprints with 80.2% confidence.

FAKE PROBABILITY	REAL PROBABILITY	VERDICT	CONFIDENCE
84.5%	15.6%	FAKE	80.2%
Softmax Output	Softmax Output	Based on 78.5% Threshold	Calibrated Score

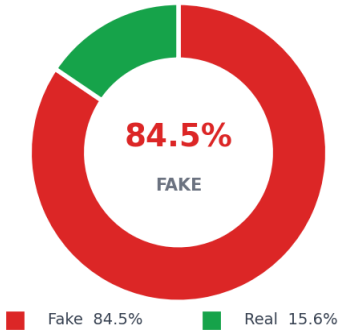
PROBABILITY ASSESSMENT (AUDIO)

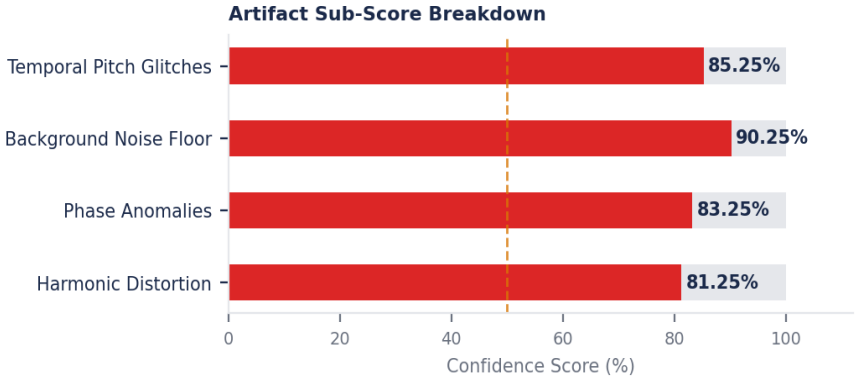


CASE INFORMATION

FIELD	VALUE
Case ID	CASE-1775337685334
Report ID	A9EDF525-A709-44
Source File	Text To Speech_1000.wav
Analysis Type	Acoustic Deepfake / Voice Synthesis Detection
Frontend Model	WavLM Base+ (Microsoft, frozen)
Classifier	Self-Attention DeepFakeDetector Head
XAI Methods	Integrated Gradients, SHAP
Audio Duration	4.037 seconds
Generated	2026-04-04 21:21:26 UTC

AUDIO ARTIFACT ANALYSIS



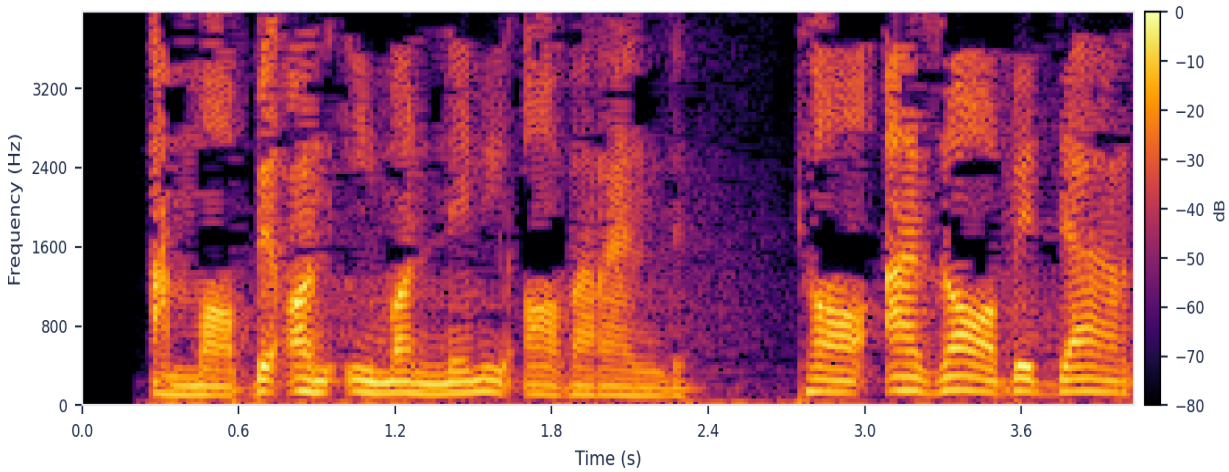


EXECUTIVE SUMMARY

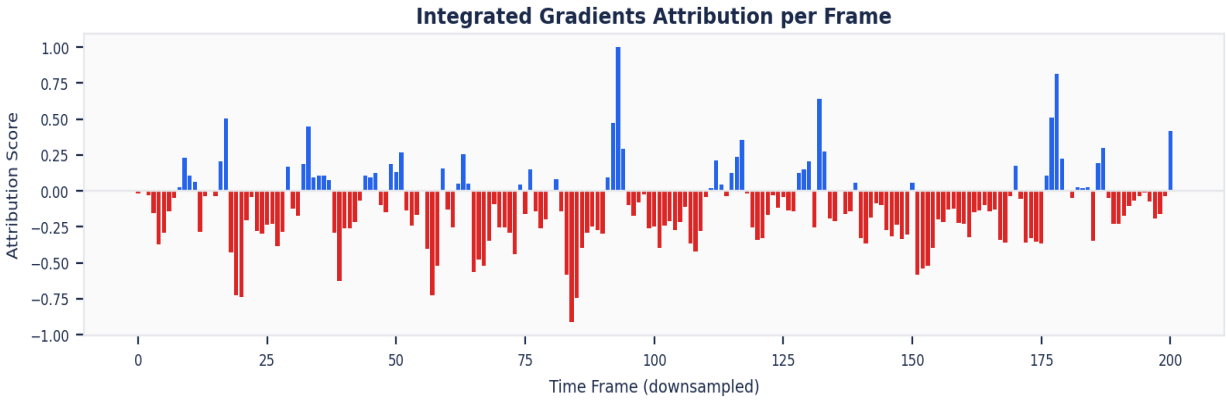
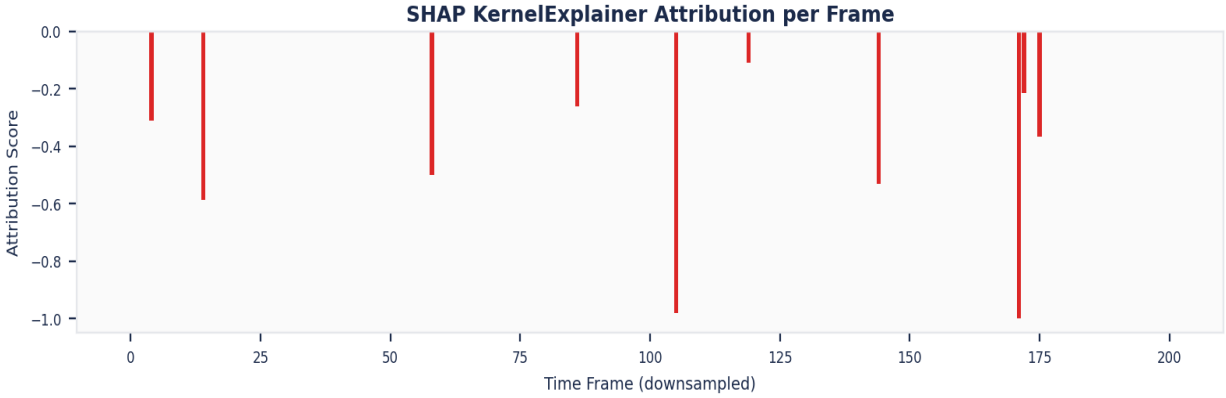
WavLM feature extractor detected significant acoustic artifacts consistent with AI-generated audio or voice cloning. The classifier output indicates a 84.5% probability of synthetic generation.

ACOUSTIC FORENSICS MAP

The Short-Time Fourier Transform (STFT) spectrogram visualises the frequency content of the audio signal over time. Artificial speech synthesised by neural TTS models exhibits characteristic spectral anomalies such as unnatural harmonics.



DETECTED ACOUSTIC ANOMALIES

ANOMALY INDICATOR	VALUE	INTERPRETATION
Synthetic Speech Prob	84.45%	HIGH — exceeds EER threshold
Authenticity Prob	15.55%	Residual probability assigned to genuine class
Audio Duration	4.037s	Actual processed duration (padded to 4s)

TEMPORAL ATTRIBUTION — IG & SHAP